

	A	B	C	D	E	F	G	H
1	Power Budget Example							
2	Team Number:	209						
3	Project Name:	Door Sensor						
4	Team Member Names:	Bryce, Matthew, Dylan						
5	Version:	0.01 Pre Release						
6								
7	A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
8	All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
9		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
10		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
11		Dynamic Speaker Unit	SP-4005Y	+2.8 - 3.4V	1	42	42	mA
12		Omni-Directional Microphone	AOM-4544P-2-R	+1.5 - 10V	1	0.5	0.5	mA
13		5V regulator	AP7375-50SA-7	+5V - 45V	2	300	600	mA
14								
15								
16	B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
17	+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
18		PSoC™ 4 BLE Module	(full part number)	(range)	1	200	200	mA
19		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
20		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
21		Dynamic Speaker Unit	SP-4005Y	+2.8 - 3.4V	1	42	42	mA
22		Omni-Directional Microphone	AOM-4544P-2-R	+1.5 - 10V	1	0.5	0.5	mA
23						Subtotal	262.67	mA
24						Safety Margin	25%	
25						Total Current Required on +5V Rail	328.3375	mA
26								
27	c2. Regulator or Source Choice	5V regulator	AP7375-50SA-7	+5V - 45V	2	300	600	mA
28						Total Remaining Current Available on +5V Rail	271.6625	mA
29	C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each							
30								
31	D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need							
32	External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
33	Power Source 1 Selection	Plug-in Wall Supply	N/A	110VAC	+24V	15000	15000	mA
34								
35	Power Rails Connected to External Power Source 1							
36		5V regulator	AP7375-50SA-7	+5V - 45V	2	300	600	mA
37								
38						Total Remaining Current Available on External Power Source 1	14400	mA
39								
40	External Power Source 2	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
41	Power Source 2 Selection	Battery	AA	+120V	1.5V	25000	25000	mA
42								
43	Power Rails Connected to External Power Source 2	5V regulator	AP7375-50SA-7	+5V - 45V	2	300	600	mA
44								
45								
46						Total Remaining Current Available on External Power Source 2	24400	mA

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47								
48	E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.							
49	Component Name	Part Number	SupplyVoltageRange	Capacity(mAh)	RequiredByRegulators			
50	Battery	AA	1.5V	500	600			
51						Battery Life	0.833333333	hours
52	Notes							
53	External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).							
54	If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit							